

The Economy

Seminar 7: Financial Markets

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Balance sheet of a bank

Assets	Liabilities
Cash	Deposits
Investments • stocks • bonds • government securities	Equity
Loans • mortgages • inter-bank loans • letters of credit	MMF
Reserves	

Accountants call:

- Your skin in the game, Assets - Liabilities: **Equity**
- Savings for a rainy day: **Reserves**

Financial assets: A partial list

- ➊ **Stocks** – ownership claims on companies, entitling owner to a share of profits
- ➋ **Bonds** – entitles owner to a steady flow of interest payments
- ➌ **Money** – cash.

Individuals hold a **portfolio** of different assets; reallocate their wealth among assets based on market returns, expectations, etc.

Monetary policy involves changing the quantity of money. Monetary, *not fiscal policy*, is actually the main tool used to fight recessions and rein in booms.

- **Monetary policy** is primarily concerned with the management of interest rates and the total supply of money in circulation and is generally carried out by central banks – ECB, FED etc.
- **Fiscal policy** is a collective term for the taxing and spending actions of governments.

To understand the role of money and monetary policy, we ignore stocks and imagine an economy in which bonds and money are the only two assets.

Think of one-year bond. You buy it for \$1, get $\$(1 + i)$ one year later.

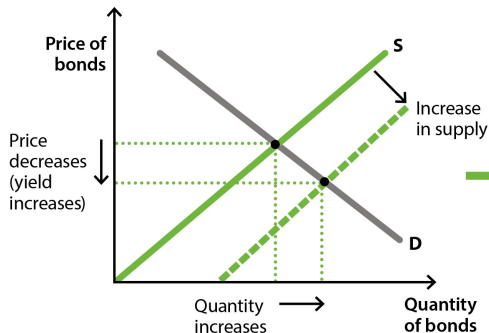
Actual bonds are often multi-period.

- E.g., you buy for \$1, receive \$ i each year for next 10 years, then get your \$1 back.
- Typical maturities range from 1 day (Fed funds rate), to 90 days (Treasury bills), to 30 years (Long term gov't bonds) or sometimes even indefinite (BoE perpetual bonds).

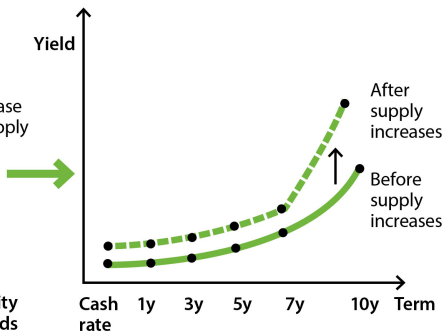
The bond market

What happens if there is news *after* a bond is issued, changing peoples' demands?
The **price** of a bond rises or falls, in order to make the effective **yield** match the market.

Ten Year Bond Market

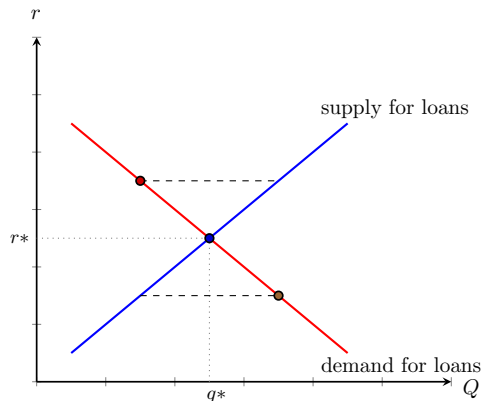


The Yield Curve



- Bond price up → interest rate down
- Bond price down → interest rate up

Loans and interest rates



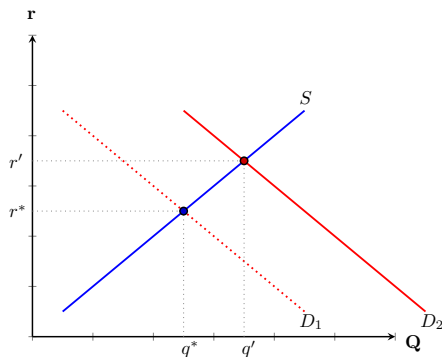
In an **OMO** (open-market operation) the CB:

- prints money to buy more bonds, putting more money into circulation
- sells bonds to withdraw money from circulation

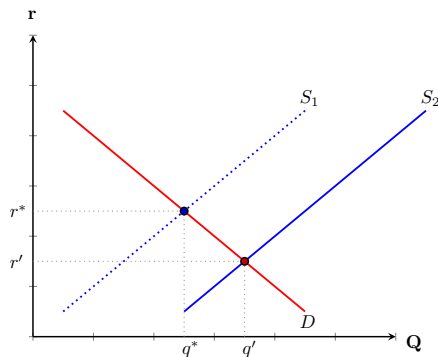
This affects the interest rate – and because the interest rate affects spending, it affects the economy as a whole.

Changes in r

The market is somewhat free to fluctuate around the price for loans (i.e. interest rates) that the CB sets. The demand side reacts to certain events and the CB has to take action accordingly to keep a delicate balance.



People wanting to invest more



ECB buying bonds

Fluctuating r

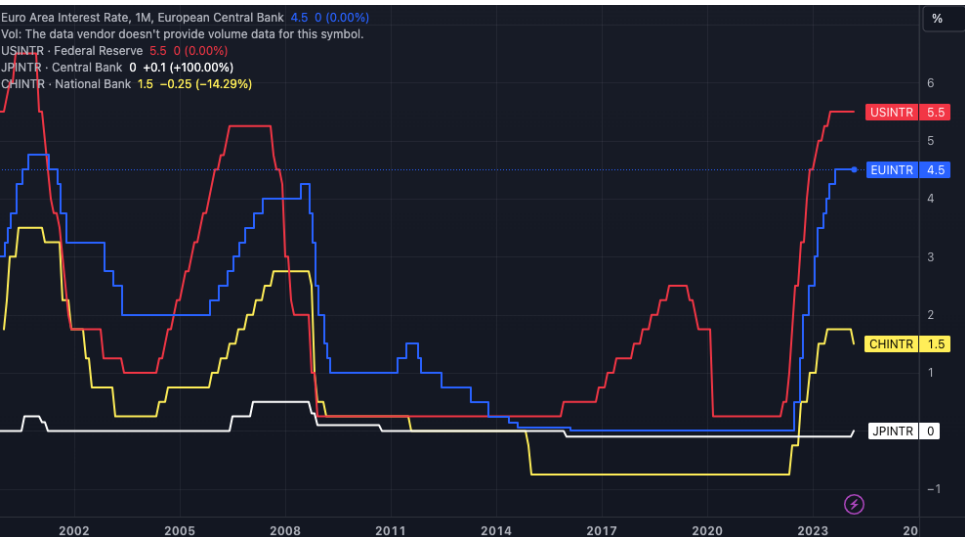
Euro Area Interest Rate, 1M, European Central Bank: 4.5 0 (0.00%)

Vol: The data vendor doesn't provide volume data for this symbol.

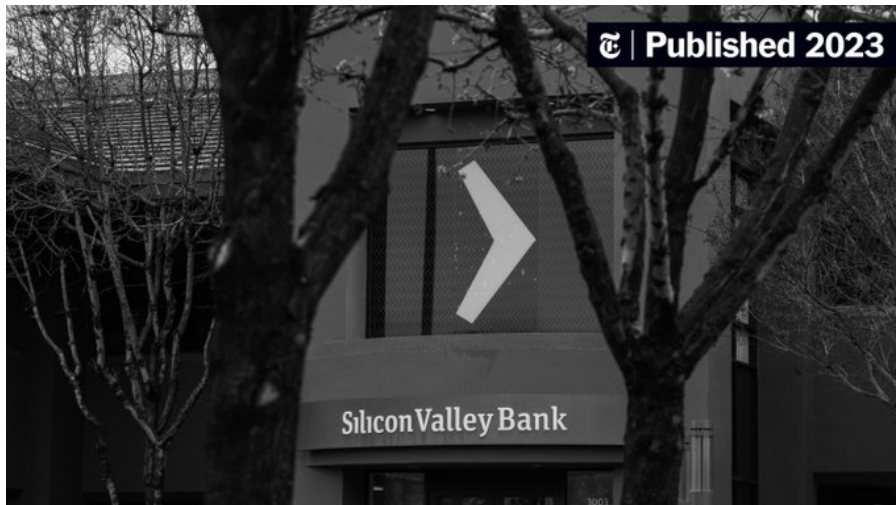
USINTR · Federal Reserve 5.5 0 (0.00%)

JPINTR · Central Bank 0 +0.1 (+100.00%)

CHINTR · National Bank 1.5 -0.25 (-14.29%)

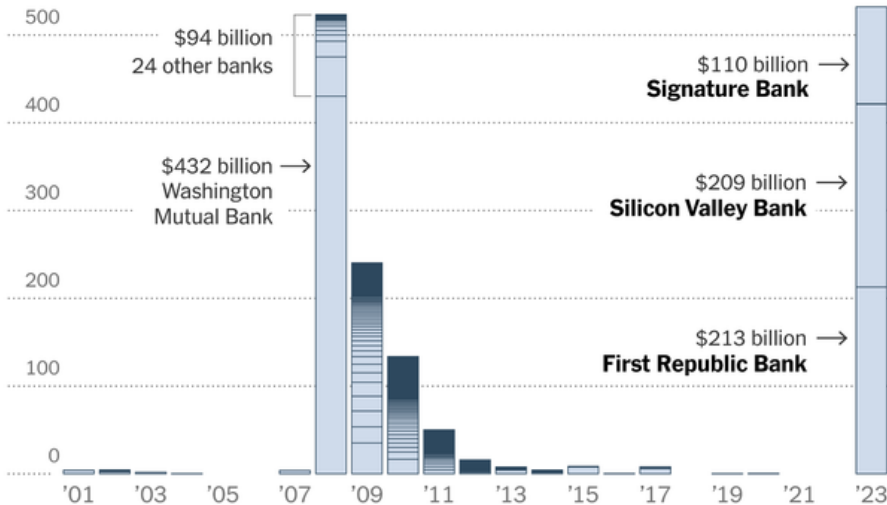


Bank runs: SVB collapse



\$600 billion

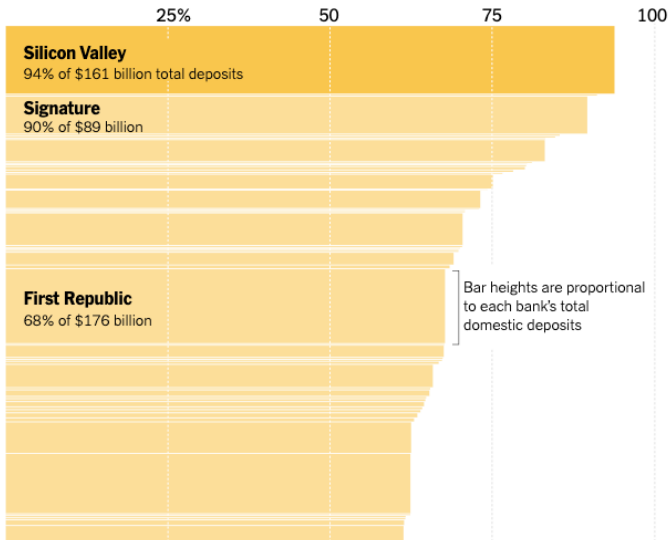
U.S. bank failures in each year, adjusted for inflation



Top 50 banks by share of deposits that are not federally insured

Excludes banking giants considered systemically important

Greater share of deposits uninsured ►



The time value of money

- The value of money changes over time. Interest rates help account for the balance between loaners and borrowers.
 - the **benefit** of putting money into the bank is the money you'd earn from it after some time
 - the **price** you pay for consuming today is the money bank charges you after some time
 - all of which is based on interest rates.
- We refer to the value of money at different points in time as **present value** and **future value**.

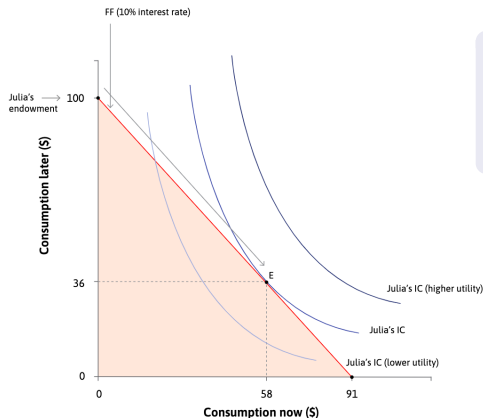
$$FV = PV(1 + r)^n$$

where n is the number of periods. Rearranging, we get:

$$PV = \frac{FV}{(1 + r)^n}$$

Borrowing and r

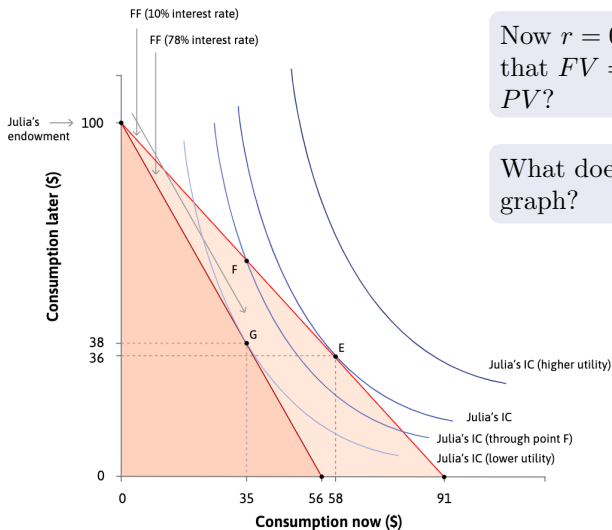
Similarly to the classic idea of budget constraints and ICs, we can represent an individual's preference between consuming now and later.



Julia has no money but she'll get \$100 in a year. Trusting on that, she can decide to borrow money today.

- when $r = 0.1$, Julia's optimal bundle is at **E**
- here, $MRS = MRT$
- Julia spends $58 + 36 = \$94$ in total

Borrowing and r



Now $r = 0.78$. Given we know that $FV = \$100$, what is the PV ?

What does it show us on the graph?

PS7 Exercise 1: Consume today vs. tomorrow

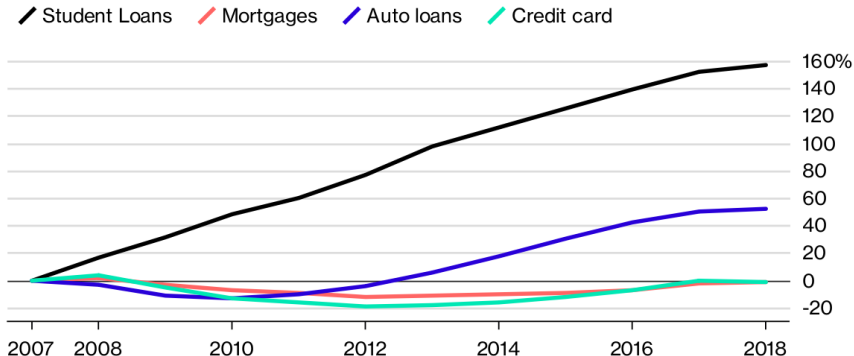
Today is Matilda's 25th birthday. She has 200€ which she wishes to use to buy fireworks for her birthday and the next one. Indeed, she only uses fireworks at a specific time of the year, for her birthday.

Moreover, she knows that she can invest her money at an interest rate of 10% per year and that the price of a firework is 20€ today as well as in one year.

- ❶ Graph the set of possible combinations of present consumption/future consumption of fireworks by Matilda.
- ❷ Draw indifference curves for Matilda and deduce her optimal present and future fireworks consumption.
- ❸ We consider the same problem, but with an interest rate of 15%. Can we say that, compared to the previous example:
 - the present consumption will necessarily be lower?
 - the future consumption will necessarily be higher?

Student Debt Just Keeps Growing

Student loans are the fastest growing segment of U.S. household debt, seeing almost 157 percent growth since the Great Recession.



Source: Bloomberg Data

Bloomberg

Unemployment

We study the labor market because it's the key to the **supply side** of the economy. When economists are studying the economy, they tend to hone in on two factors: output and jobs. Because there is a relationship between these two elements of an economy, many economists study the relationship between GDP and unemployment levels.

Reminder: Unemployment is the ratio of people looking for jobs but not employed, to the total work force (unemployed + employed).

Unemployment,

- changes dramatically over the **business cycle**
- is closely associated between changes in unemployment and growth (Okun's Law)

Okun's Law

Okun's Law is an empirically observed relationship between unemployment and losses in a country's production. It predicts that a 1% increase in unemployment will usually be associated with a 2% drop in the GDP.

Wage determination

In a free market, the wages are determined by the quantity of the labor supply and the demanded labor.

- People care about their wage or real wage $\frac{W}{P}$ (i.e. how much you can actually buy with your wage)
- Workers have more bargaining power, employers are more anxious to pay high “efficiency wages” when unemployment is low
- Most wages are fixed for some length of time (a year or more)

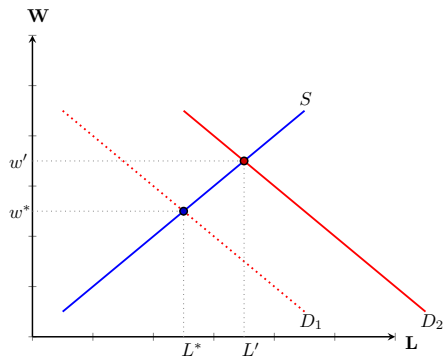
Then, the implied wage equation would be:

$$W = P^e F(u, z)$$

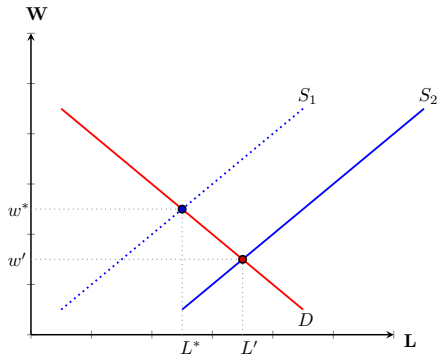
- P^e — expected price level. Price level because workers care about real wage. Expected because wages must be set in advance.
- u — unemployment rate. Negatively correlated with w because of bargaining, efficiency wages
- z — all other factors

Changes in w

Just like the market for loans and goods, the shifts in demand and supply affect the equilibrium (i.e. w^*)



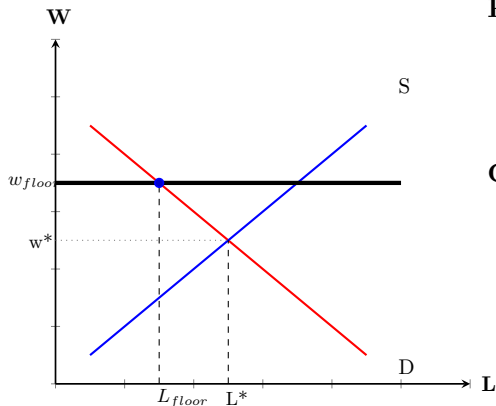
Job creation



Immigration

Minimum wages

A minimum wage is the lowest wage that a worker may be paid per hour. It's a price floor on hourly wages mandated by federal law. Workers can't be offered a job or agree to work for less than this amount.



Pros:

- better standard of living
- increased employee morale
→ increased productivity
- increases consumption investments

Cons:

- reduces the demand for labor in the economy
- increasing prices overall
→ inflation
- increases the portion of jobs with a minimum wage

Beveridge curves

Beveridge curves show the relationship between **unemployment** and the **job vacancy rate**, the number of unfilled jobs expressed as a proportion of the labor force.

- it has a **downwards slope**:
a lower rate of job vacancy $\rightarrow$ higher rate of unemployment
- outward movements over time suggest that given level of vacancies would be associated with higher and higher levels of unemployment
 $\rightarrow$ inefficient labor market.

This is caused by mismatches between available jobs and the unemployed and an immobile labor force

Series Id: LNS12300060
Seasonally Adjusted
Series title: (Seas) Employment-Population Ratio - 25-54 yrs.
Labor force status: Employment-population ratio
Type of data: Percent or rate
Age: 25 to 54 years

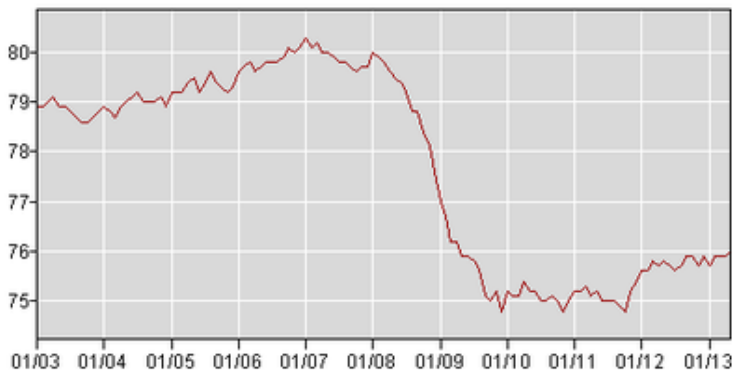
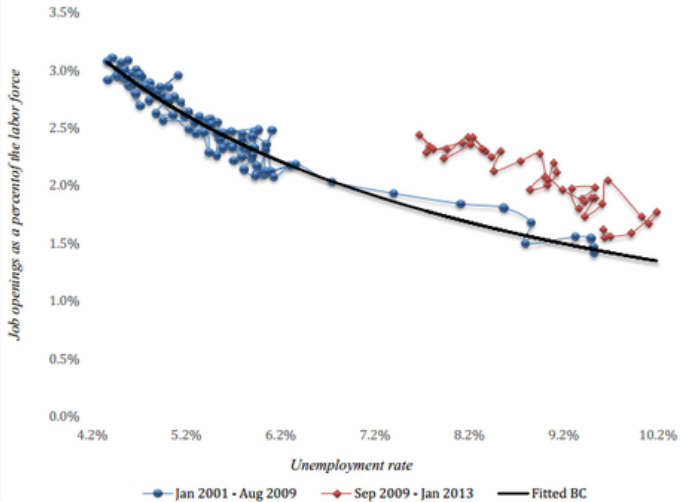


Figure 1: Monthly Vacancy and Unemployment Rates
Jan. 2001 – Jan. 2013



PS7 Exercise 2: Technology and the labor market

- ④ Consider a point on the Beveridge curve for the year 1981. How does unemployment evolve after a negative shock to aggregate demand? According to the Beveridge curve, how do you expect the vacancy rate to move?

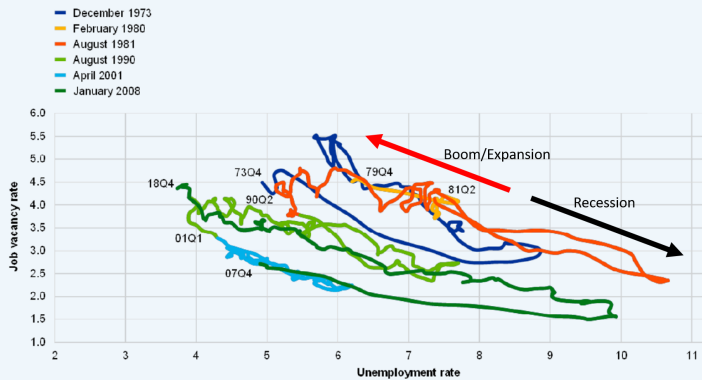
- a fall in aggregate demand means less revenue, company has to hire less and fire more, so unemployment increases and the vacancy rate decreases
- we move in the south-east direction of the Beveridge curve

PS7 Exercise 2: Technology and the labor market

- ② Visualize a plausible pair of unemployment/vacancy rates for the year 1982 as a point on the graph. Mark two directed arrows describing the evolution of the unemployment and vacancy rates during a boom or a bust.

The US Beveridge curve

(as a percentage of the civilian labour force; three-month moving averages)



PS7 Exercise 2: Technology and the labor market

- ③ Compare the Beveridge curve for 2001 (light-blue) with the Beveridge curve of 1981. Describe in words the differences between the two curves. Which Beveridge curve features a higher unemployment rate for a given vacancy rate? Do the two Beveridge curves differ their elasticity of unemployment with respect to the vacancy rate?
- the red 1981 BC and the light blue 2001 BC have similar slope but the 2001 BC is closer to the origin
 - this means that for a given vacancy rate - there is a higher unemployment in 1981 compared to 2001
 - the elasticity of unemployment with vacancy (essentially the slope here since the axes are in %-terms) has not changed much

PS7 Exercise 2: Technology and the labor market

- ④ Using your previous result, what can you say about the relative efficiency of matching (unemployed) workers to firms in the US labor market in the 1980s compared to the early 2000s ? Name two types of structural change that could have plausibly caused this shift.
- since the 2001 BC is closer to the origin, for any u -level, there are fewer vacancies and for any v -level, there is lower unemployment! the market does a better job at matching the unemployed with vacancies, and is thus more efficient
 - a BC moves closer the origin when matching is easier and skill compatibility is higher between job seeker and vacancy
 - the introduction of internet facilitated the matching, an increased access to education to match the arrival of the internet facilitated skill compatibility, further urbanization and connectivity of the urban space allowed for better matching, liberalization of the labor market allowed more matching

Next...

- **PS8 Exercise 1** due in class.
- No quiz next week.
- Final exam on 6th of May.